telecom

MDPI

Review

Blockchain Interoperability for Next Generation Telecommunication Revolution

Gagik Sarkisyan [1,*], Gören Ramadov [2] and Tacha Saninyan [1]

check for updates

Academic Editor: Maurizio Pizzonia

Received: 07 June 2025
Revised: 11 July 2025
Accepted: 20 July 2025
Published: 27 July 2025

Citation: Sarkisyan, G.; Gören, R.; Saninyan, T. Blockchain Interoperability for Future Telecoms. *Telecom* **2025**, *6*, 20. https://doi.org/10.1190/telecom6010020

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license

[1] Computer Engineering, Yeditepe University, Istanbul 34755, Türkiye; tsaninyan@cse.yeditepe.edu.tr
[2] Electrical and Computer Engineering, University of Massachusetts Dartmouth, North Dartmouth, MA 02747, USA; ramadov.goren@umassd.edu
* Correspondence: tsaninyan@cse.yeditepe.edu.tr

Abstract: Blockchain's unique properties—such as immutability, autonomous execution, and the elimination of intermediaries—continue to drive significant interest in its potential applications for the telecom industry, making it a compelling area for ongoing investment. This literature review focuses on the emerging and impactful domain of blockchain interoperability, which aims to enable information exchange between distinct, independent blockchain networks. By harnessing interoperability capabilities, telecom operators can facilitate seamless data sharing with other mobile, fixed, and next-generation networks, resulting in greater security, operational efficiency, cost reduction, and enhanced user experiences. This work critically analyzes leading research papers related to blockchain's use cases for interoperability in telecom, introduces key interoperability solutions, and outlines the fundamental requirements needed for successful implementation. The review also identifies major gaps in current research and suggests directions for future developments, particularly as the sector transitions toward sixth-generation (6G) telecommunications.

Keywords: blockchain; Hyperledger; interoperability; permissioned; SAGIN; telecommunications; 6G

1. Introduction

With the rise of sixth-generation (6G) networks and innovative cloud-based, open, and flexible self-executing telecom solutions such as open radio access network (O-RAN) [1–3], telecom networks are expected to become more adaptable and capable of automatically extending their capacity for much more extensive use cases. The Internet of Everything (IoE) is expected to be fully realized in the 6G context, bringing many next-generation devices to networks. Such a flexible and complex architecture demands robust security and access features. In this context, blockchain is anticipated to play a crucial role in the future of telecoms. Furthermore, blockchain can enhance secure interoperability in telecommunications by enabling various telecom networks and systems to connect and exchange data effortlessly. According to Georgios et al. [4], while fifth-generation (5G) networks have enabled the IoE, 6G will tackle the numerous limitations of 5G and further improve the Internet of Things (IoT). This is anticipated to result in a vast increase in the number of devices connected to mobile networks. Such growth has significant potential to introduce major security challenges in 6G. Additionally, the requirement for the seamless access of these IoE devices to other networks as they move through the air, sea, and space poses further challenges.Ramraj et al. [5] contend that various technologies—including quantum computing.

This literature review is novel in its focus on identifying the essential characteristics of blockchain interoperability for future telecom use cases and challenges. The identified use cases and associated challenges in the context of 6G primarily pertain to the domains of robust security, privacy, and real-time execution. These factors are critical to ensuring that dynamic resource allocation and real-time interconnection are effectively managed and optimized. It also presents existing interoperability solutions by introducing a new taxonomy called the side-by-side approach, which meets these important criteria. This new approach can be regarded as a promising option for future research.

1.1. Problem Statement and Motivation

Most research into blockchain use cases in future telecom technologies focuses on deploying a single blockchain and its interoperability with internal applications through decentralized applications (dApps). According to Ren et al. [6], blockchain interoperability operates at three distinct levels:

- A: among blockchain and other systems;
- B: among dApps on the same blockchain;
- C: among different blockchains.

In Figure 1, the three levels of blockchain interoperability are illustrated according to the description of Ren et al. [6]. Levels A and B are integral to research projects focused on existing and future telecom technologies. Studies such as [7] exemplify research projects that demonstrate interest in using blockchain in telecoms. In [7], the authors proposed dynamic spectrum allocation in 6G using a Level A oracle [8,9], which they call an intermediary. Additionally, they covered a use case for the interoperability of Level B using dApps [10,11] in the same research project. However, there is limited research and almost no studies on the interoperability of independent blockchains (Level C) operating across different telecom networks. This lack of research motivated this literature review to explore the Level C type of blockchain interoperability and its use cases in the context of telecommunications.

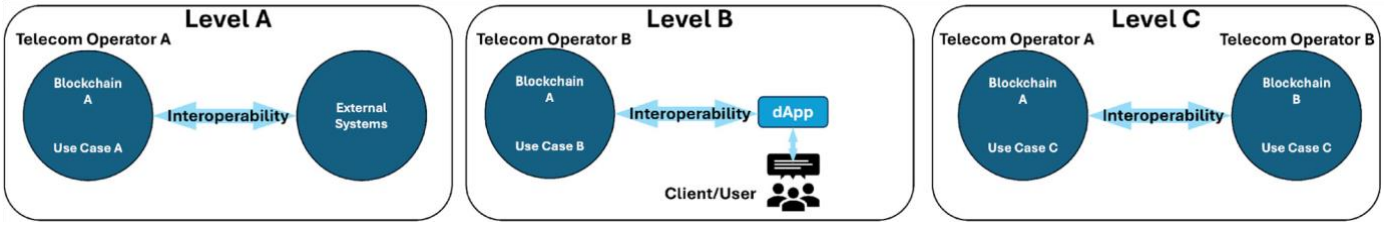

Figure 1. Three levels of blockchain interoperability in telecom. dApp, decentralized application.

With the introduction of 6G, a significant number of edge devices are anticipated to be deployed in mobile networks, which raises issues relating to security, privacy, and scalability. The deployment of multi-access edge computing (MEC) nodes allows for the processing of sensitive data at the network edge (e.g., federated learning), thereby increasing vulnerability to data tampering, DDoS attacks, Sybil attacks, and insider threats. Traditional centralized identity management systems are prone to single points of failure (SPoFs), data breaches, and identity spoofing in large-scale IoT and MEC deployments. Self-sovereign identity (SSI) deployed with blockchain allows devices, users, and network components to be authenticated using decentralized identity (DID) mechanisms stored on a blockchain, thereby preventing unauthorized access.

Along with significant improvements in latency and reliability, 6G introduces AIdriven network optimization, seamless interconnection, and ubiquitous connectivity through the space–air–ground integrated network (SAGIN) concept. The novel features and their challenges can be effectively addressed by leveraging blockchain technology and its interoperability capabilities. The

primary aim of this literature review is to emphasize the critical aspects of this research area and propose future directions, particularly for Level C-type interoperability use cases. Through the use of a blockchain interoperability architecture—particularly for Level C use cases—telecommunications providers can tackle the challenges of security, privacy, fragmented data, and interoperability issues with their external partners, including roaming, interconnection, and ecosystem partners. In current telecom practices, employing trusted third parties (TTPs) is a common method whenever data exchange between two or more telecom operators is required.

On the other hand, TTPs are vulnerable to security issues, censorship, and privacy concerns. Blockchain technology can facilitate reliable peer-to-peer transactions without compromising security or privacy by removing the need for trusted intermediaries. Utilizing a blockchain interoperability architecture allows telecommunications providers to unlock new opportunities for securely and efficiently connecting and exchanging information between diverse networks and systems. Another emerging area of interoperability research is [12,13], which is becoming a vital part of decentralized organizations and is increasingly recognized as an essential component of the Metaverse [14]. Assuming that both Web3 and the Metaverse will be part of future telecom networks, the significance of this literature review becomes much more critical for researchers.

1.2. Structure of the Manuscript

This study begins by demonstrating the history of blockchain technologies by highlighting critical technological improvements in the blockchain domain, including significant milestones that new blockchain technologies have achieved. Then, it illustrates how blockchain is becoming an essential technology for enterprise use cases by introducing interoperability. Next, it explains why telecoms require blockchain interoperability (i.e., cross-chain communication). Critical telecom interoperability use cases that significantly influence the industry's future, including their benefits and challenges, are also illustrated. Key research papers are reviewed to provide more precise guidance for researchers in this field, highlighting essential features that can be considered crucial in selecting the ideal solutions for telecom use cases. Next, the chosen solutions are compared to assess their alignment with the highlighted features. In conclusion, a comparison table illustrates the most effective solutions based on the feature set and the findings of the literature review. The research contributions are also outlined, and key topics and challenges for future discussions are highlighted.

This literature review concludes that the research area of telecom interoperability is still in its early stages, especially concerning future 6G deployments.

1.3. History of Blockchain and Interoperability

Blockchain networks were initially designed to function as standalone decentralized solutions. A decentralized approach eliminates trusted parties and provides an automatic and immutable system for proof of transactions. The idea of blockchain was first offered by the inventor of Bitcoin, Satoshi Nakamoto [15], as a decentralized, public, permissionless, standalone, hashed block-based cryptographic proof system that eliminates trusted parties in the context of payment verification. There are two main types of blockchain systems: permissionless and permissioned. Permissionless blockchains operate as public networks and allow members to join without requiring approval. This design approach plays a significant role in decentralized blockchain solutions and serves as a reference blueprint for blockchain-based cryptocurrency solutions. As public permissionless blockchain designs initially focused on cryptocurrency and standalone networks, the demand for general blockchain interoperability was minimal until new popular cryptocurrencies, such as Ethereum [16], were launched. Meanwhile, three well-known challenges have emerged in this research area: security, flexibility in adaptation, and performance.

According to [17], the Bitcoin consensus algorithm is a widely debated topic in the Bitcoin research community due to its stability, security, and concerns regarding computational resource

wastage. Ethereum was developed and launched by Buterin et al. [10] in 2015 to address some of these challenges. It aimed to provide flexibility to cryptocurrency networks through the introduction of smart contracts [10], which enable more adaptable transaction processing. The introduction of smart contracts has facilitated the integration of enterprise rules and application usage. Through smart contracts, new enterprise-based features can be added to the current consensus mechanism, such as proof of work (PoW) and proof of stake (PoS). Finally, a new concept called decentralized applications (dApps) enables public blockchain networks to interact flexibly with other applications. dApps were introduced within Ethereum projects, allowing for interactions with centralized and decentralized external applications using smart contracts. The introduction of smart contracts and dApps has also facilitated extended use cases beyond cryptocurrency deployments and has become a significant use case for various industries. Although Ethereum is a more flexible solution, Yakovenko proposed Solana [17] in 2016 to provide more high-performance blockchain-based cryptocurrency networks to address well-known performance challenges. With the effect of the newly launched cryptocurrency networks, interoperability projects have emerged for public cryptocurrency-based blockchains. Such projects were primarily built to meet the growing demand for transferring or swapping cryptocurrency assets between networks. According to [18], over 10,000 blockchain networks had been reached by 2022, and significant amounts are still being built for public cryptocurrency projects.

New cryptocurrency-based interoperability projects, such as Cosmos [19], Polkadot [20], and Overledger [21], have been launched to address asset and coin transfers and swaps. These projects were deployed as asset or coin transfers, which can only be achieved through interoperability, and aim to advance cross-industry blockchain technologies for enterprise use cases.

In 2016, the Linux Foundation launched the Hyperledger project with 30 founding members in order to advance cross-industry blockchain technologies for enterprise use cases. The objective was to create an open-source framework for building distributed ledger solutions beyond cryptocurrencies. The Hyperledger project facilitated the enterprise-wide usage of blockchain, particularly for private, secure, permissioned networks. This initiative also prompted discussions on interoperability projects for industry and enterprise use, which is discussed in this article. This literature review primarily focuses on cases of blockchain interoperability that apply to enterprises and their relevance to future telecom requirements. This focus eliminates the concentration on cryptocurrency-based permissionless use cases that are less relevant to enterprise applications. As industries undergo digitalization and enterprises experience digital transformation, blockchain technologies and interoperable blockchain architecture frameworks can address several challenges in the industry. Before discussing the details of interoperability, it is necessary to explore how it can reshape the telecommunications industry and what benefits it can add.

2. Study Background

2.1. Interoperability Requirements in Telecommunications

In the telecommunications industry, there is a growing need to enable seamless, secure, and tamper-proof transactions for decentralized decision-making and data exchanges between service providers. A straightforward way to address this need is to use blockchain and cross-chain communications. However, achieving this cross-chain communication in the telecommunications infrastructure poses challenges, including technical complexities, particularly relating to interoperability, scalability issues, and regulatory hurdles to data privacy and security. According to Jabbar et al. [22], in order to address these challenges, telecommunications companies and blockchain developers need to collaborate and establish common standards for interoperability. High-quality surveys, literature reviews, and design principles are critical for developing these standards. Furthermore, implementing interoperability for telecommunications requires addressing the fragmentation of data, reliable provenance, and diverse protocol regulations across

different distributions and processes. To address and overcome these challenges, it is essential to identify which telecom use cases best suit blockchain interoperability.

2.2. Blockchain Interoperability Use Cases in Telecommunications

Telecommunications, a heavily regulated sector with high-performance expectations, is still in the early stages of realizing use cases for blockchain interoperability. One critical benefit of blockchain and its interoperability needs is its ability to make decentralized decisions. Below are the predominant applications in the research domain that require decentralized decision-making processes.

2.2.1. Identity Management

Due to stringent regulatory and privacy requirements, telecom operators must ensure user privacy and security. Therefore, telecom companies are actively exploring blockchainbased identity management solutions. These solutions are anticipated to enhance security and privacy while enabling secure interoperability among various telecom service providers. According to [23], cloud-based cognitive networks require more comprehensive identity management when users seek access to multiple operators. Implementing dynamic spectrum allocation in next-generation 5G and 6G networks requires secure identity management to enable seamless switching between different spectra. Using secure, immutable blockchain technologies enables telecom operators to empower users to maintain control over their identity data and streamline authentication processes across various telecom networks. This authentication process requires secure data exchange between operators. Given regulatory restrictions, whether two or more independent telecom operators can join a single blockchain network and share the same infrastructure is a valid point of discussion. Consequently, implementing one identity solution for multiple networks and consolidating operator user bases into a single platform presents significant regulatory challenges. Due to strict regulatory requirements and data privacy rules, such as the General Data Protection Regulation (GDPR) in the European Union, telecom operators are forced to implement isolated blockchain networks. Secure data exchange layers (cross-chain communication) that provide interoperability with other networks are potential candidates to overcome this challenge. Such an architectural approach can be a standard deployment model for telecom use cases. Once such a secure architecture is deployed, decentralized decision-making can be enabled for many telecommunications use cases, satisfying regulatory hurdles.

2.2.2. Sixth-Generation Mobile Networks

With the evolution of 6G, many research papers and surveys, such as [24,25], have aimed to address the new opportunities that blockchain technologies provide. A comprehensive survey by Saravanan et al. [23] demonstrated the essential use cases, technical aspects, and challenges of blockchain technologies for 6G technology. Saravanan et al. discussed the challenges of deploying blockchain technologies, including their interoperability. According to [23], interoperability is frequently mentioned as a challenge and a topic of open discussion regarding telecom use cases. According to [24], using blockchain for 3D networking in 6G can provide decentralized and secure data access, offloading, and interoperability solutions. Global coverage is a significant feature of cellular network connectivity in 6G. Based on the initial requirements [26] of 6G mobile networks, three-dimensional (3D) global coverage has been envisioned. The international communication coverage in 6G encompasses the air, space, underground space, and sea. Such coverage requires continuous interactions and strictly controlled access requirements between the layers and other networks. In this sense, smart contracts, distributed ledger technologies, secure identity management, and interoperability introduced by blockchain are strong candidates. These technical requirements are significant features that allow seamless 6G access and coverage [24]. They also facilitate further use cases, such as Industry 5.0, Smart Healthcare, unmanned aerial vehicle (UAV)

applications, connected autonomous vehicles (CAVs), Energy Internet, Extended Reality (XR), Holographic Telepresence, and Smart Cities.

Figure 2 depicts the 6G SAGIN [27] architecture, which is expected to be deployed in 6G mobile networks by 2030. Future directions, opportunities, and challenges for deploying seamless, integrated, and converged architectures have been discussed in [28]. The following sections discuss how blockchain interoperability can be deployed through the use of SAGIN [27] architectures.

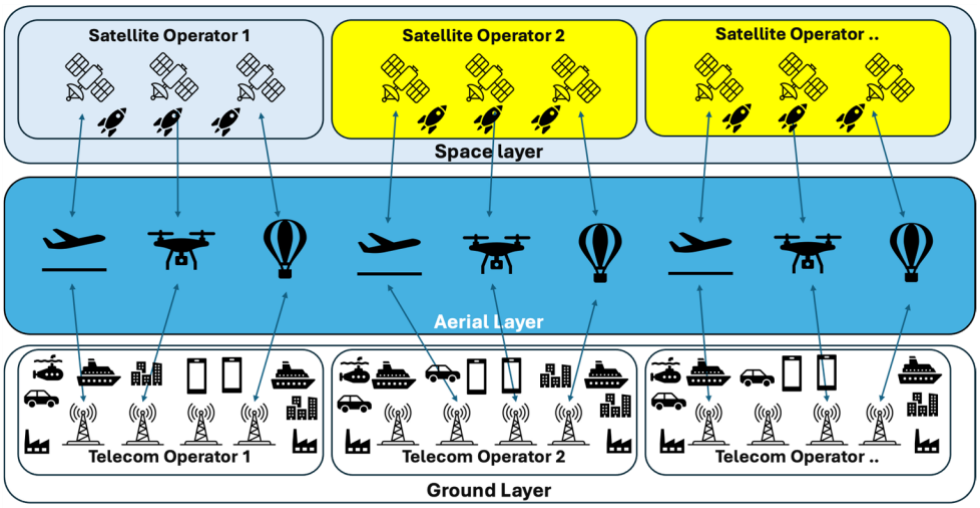

Figure 2. Sixth-generation (6G) space–air–ground integrated network (SAGIN) architecture.

2.2.3. Roaming Services

Roaming services have become a critical requirement in deploying mobile communication services. If a roaming agreement is signed between the home country and the mobile operators of a foreign country, mobile users can enjoy mobile services while visiting that country. Based on roaming agreements, a visitor's use of services and usage records are submitted to a clearing house—that is, an intermediary between the operators. In this process, the clearing house applies processing and settlement fees, resulting in higher costs for the end users. Figure 3 illustrates international roaming between two mobile operators in existing mobile networks.

Decentralized interoperable blockchain platforms can optimize roaming services through providing transparent, intermediary-free, and real-time settlement mechanisms between telecom operators. A candidate blockchain solution can use smart contracts as preset roaming exchange rules and automatically apply the process if a roaming service is triggered in the visiting country. Finally, it calculates the roaming fees without an intermediary, such as a roaming clearing house. Salah et al. [29] discussed roaming as an opportunity in 5G, which can be addressed using blockchain. They discussed international and national roaming, in which blockchain can improve the quality and experience of roaming services. According to Salah et al., blockchains help to reduce costs and fraud, improve billing accuracy, and enhance the overall user experience of international and national customers. The Global System for Mobile Communications Association (GSMA) [30] has already developed the "GSMA eBusiness Network" [31] using a private, permissioned, industry-wide blockchain network. This blockchain solution focuses on wholesale roaming, clearing, and settlement as the industry's first application. With the introduction of the 6G SAGIN architecture, roaming services are expected to become more complicated and subject to privacy leakage if privacy and identity requirements are not considered. According to [24], roaming fraud is expected in 6G networks due to their increased complexity and interoperability requirements.

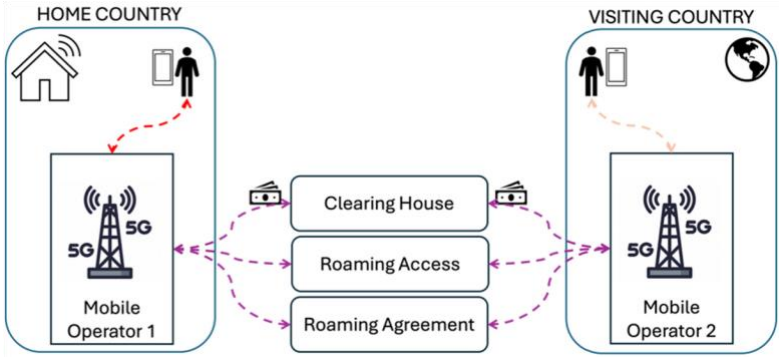

Figure 3. International roaming records processing flow for mobile networks.

2.2.4. Mobile Number Portability

Mobile number portability (MNP) is primarily deployed through intermediaries, such as roaming. In existing 5G and previous-generation networks, a clearing house is an intermediary in each country, where MNP is deployed as a semi-automated or fully automated process. Once regulatory requirements have been established, such a process can be efficiently designed using smart contracts as a self-executing autonomous process. Blockchain can easily facilitate number portability by enabling the secure and efficient transfer of phone numbers between different telecom operators using smart contracts and distributed immutable ledgers provided by blockchain technologies. With the addition of interoperability features, blockchain networks can ensure seamless porting processes while maintaining data integrity and security. In [32], Krishnaswamy et al. offered an MNP solution using a permissioned private blockchain based on Hyperledger and smart contracts. According to [32], automated, immutable processes with smart contracts increase trust, reduce the MNP processing time, and eliminate processing errors through a transparent, fully automated, and distributed process. The traditional mobile number portability process is illustrated in Figure 4.

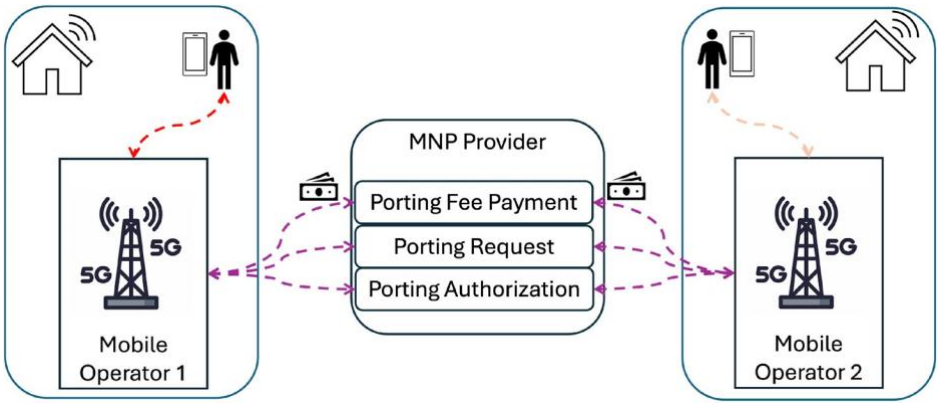

Figure 4. Mobile number portability (MNP) porting process.

2.2.5. Supply Chain Management

According to Li et al. [33], dynamic customer requirements for existing and new services force telecom operators to manage and track their existing supply chains more effectively. In this respect, they claimed that well-known features such as decentralization and openness in data tracking and management are clear benefits regarding the use of blockchain in supply chain tracking. According to Li et al., blockchain offers a tamper-proof architecture when used for supply chain tracking. Li et al. proposed utilizing blockchain for comprehensive supply tracking and management when implementing the process across multiple vendors and suppliers, including financing and logistics companies. Once the entire process is executed and tracked in the blockchain, it can be considered reliable proof of its existence. Telecom companies can leverage blockchain interoperability to manage their supply chains more efficiently by connecting to other

blockchain networks outside their organizations. Through integrating blockchain networks among their vendors and partners, telecom firms can enhance the transparency, traceability, and efficiency of the supply chain process accountability.

2.2.6. Sharing and Monetization

According to a Deloitte Insights report [34], telecom companies have a unique position in managing the relationship between content and customers. Third parties provide most of the content offered to telecom networks, thus enhancing the creation and provision of new content by third parties. This surge in content creation poses challenges for telecom service providers as they consistently strive to maintain a stable network. The new requirements introduced by the new content can be challenging to manage, as they aim to avoid frequently introducing new features into a stable telecom environment network. Blockchain can effectively manage all existing and new requirements, with a process flow built using smart contracts and an autonomous consensus mechanism. Blockchain interoperability has a high potential to enable telecom operators to share and monetize their data assets securely with third-party service providers, advertisers, and researchers. In this context, interoperable blockchain platforms guarantee data privacy, integrity, and consent management while enabling seamless data exchange across networks and third-party content providers.

2.2.7. End-to-End Orchestration of 6G 3D Wireless Networks

According to [26], end-to-end orchestration is crucial in next-generation telecom networks due to the heightened expectations for low latency and seamless connectivity among multiple service providers. Low latency is a primary feature of 5G and is anticipated to become even more significant among the various endpoints associated with different service providers in 6G. The seamless connectivity in the 6G scenario requires an autonomous and fully automated process. The transparency and traceability of such processes are necessary for fine-grained access policies. In this case, through the use of an integrated interoperability architecture, blockchain can meet essential requirements and provide ideal tracing capabilities.

2.2.8. Integrated Sensing and Communications

One of the key enablers of next-generation wireless networks is integrated sensing and communications (ISAC) features, which enable the seamless connection of billions of devices. ISAC is expected to support various new 6G use cases, such as autonomous driving, human activity sensing, connected autonomous systems, and unmanned aerial vehicles (UAVs), which require massive connectivity, low latency, and high reliability. According to Chafii et al. [35], 6G presents twelve scientific challenges that necessitate revisiting the foundations of communications theory. Chafii et al. have also identified super-resolution theory [36] as one of these challenges, which is crucial for addressing new use cases in 6G communications related to ISAC. As ISAC enables simultaneous sensing and communication, security has become a key issue regarding eavesdropping and the unauthorized sensing of illegal devices accessing the uplink and downlink to transmit information signals. Bazzi, Ahmad et al. [37] present the opportunities afforded by the upper Mid-Band Spectrum for the sixth generation of wireless communications (6G). The study primarily investigates the Frequency Range 3 (FR3) spectrum, spanning from 7.125 to 24.25 GHz, as a crucial enabler for 6G communications. However, this new spectrum opportunity for integrated sensing and communication (ISAC) presents notable challenges, particularly concerning the coordination of access among a substantial number of devices operated by various network operators. The sharing of the FR3 multi-band spectrum necessitates enhanced coordination between diverse network operators and incumbent systems. Consequently, these access requirements are expected to introduce new security and privacy challenges for ISAC. In [38], a secure solution was proposed for ISAC that addresses illegal access to the transmitted signals on the up/downlink. Zhu et al. [39] mentioned that blockchain is a promising solution for secure ISAC

in 6G, allowing for an immutable, traceable, and secure approach that provides end-to-end protection and data integrity.

For ISAC architectures, blockchain can enhance security at the network and application layers through the introduction of decentralized identity (DID) [40,41] and zero-knowledge proof (ZKP) mechanisms.

2.2.9. Network Slicing and Dynamic Resource Sharing

Mobile network capacity maximization is one of the new features introduced with 5G. For critical 5G use cases such as remote surgery and autonomous vehicles, low latency and high throughput are essential. In 6G, both low latency and high throughput are expected to be further enhanced. To ensure high-quality service, the full availability of reserved network slices and spectra for these critical services must be guaranteed and well secured. Each reserved slice in a mobile network can serve various purposes, including massive connectivity, high bandwidth, low latency, and high reliability. This requires robust management of the spectrum and orchestration of network slices, incorporating immutability and traceability features provided by blockchain capabilities. In 6G, resource and spectrum usage is further enhanced, enabling the capacity of mobile operators to be utilized by other operators. This feature necessitates stringent security and interconnection requirements, which can be fulfilled through blockchain interoperability. Figure 5 illustrates a typical architecture used for network slicing and dynamic spectrum allocation. The dynamic spectrum management component records and manages spectrum availability and assigns the available spectrum based on requirements from network slicing acquirers, such as mobile network operators (MNOs), mobile virtual network operators (MVNOs), and over-the-top (OTT) providers. An intermediary broker is used between the mobile operator and custom or blockchain-based solutions. Single intermediary-based solutions are considered an SPoF design, which is serial in nature and ineffective with respect to process usage.

In [42], surveys on blockchain-enabled dynamic spectrum allocation (DSA) and slicing management were reviewed. Additionally, the authors of [42] evaluated various blockchain platforms to determine whether they can meet the performance expectations of 6G networks. They also focused on a single-chain architecture, which acts as an intermediary between the mobile operator and external parties; however, this architecture represents clear serialization. Through utilizing multiple interoperable blockchains, enhanced parallelization can be achieved.

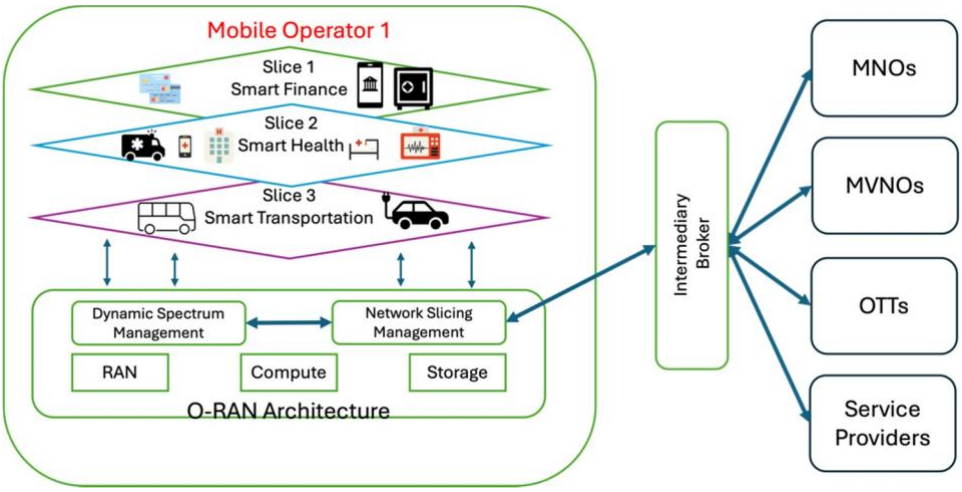

Figure 5. Network slicing and dynamic spectrum sharing architecture. O-RAN, open radio access network; MNO, mobile network operator; MVNO, mobile virtual network operator; OTT, over-the-top.

2.3. Benefits of Blockchain Interoperability in Telecommunications

The following benefits provided by blockchain can be considered the most critical:

- Enhanced security: Blockchain technology can provide a secure platform for telecommunication transactions by encrypting data and ensuring their integrity through the use of a tamper-proof hashing mechanism. DID management can be utilized alongside a ZKP approach to ensure secure authentication, providing protection against eavesdroppers and man-in-the-middle attacks. The implementation of smart contracts enhances and allows for the automation of security policies across various components of the mobile network, such as ISAC.
- Improved interoperability: Blockchain technology allows for seamless data transfer and communication between telecommunication networks, enabling better integration and collaboration among service providers with decentralized decision-making using smart contracts.
- High transparency: Integrated distributed ledger technologies make all transactions traceable and transparent to the parties involved when necessary.
- Increased operational efficiency: Blockchain technology can streamline and automate telecom processes, including supply chain management, by reducing manual intervention and improving operational efficiency.
- Cost efficiency: Blockchain technology can lower costs for telecommunications providers and their end users by eliminating the need for intermediaries and reducing transactional friction.
- Enhanced customer experience: Blockchain interoperability can enable seamless roaming services for customers, allowing them to connect to different networks, particularly 6G 3D communications, without disruptions or complicated billing processes.
- Facilitated billing and settlement: Blockchain technology, with its immutable and fully transparent features, can enable more efficient and accurate billing and settlement processes in the telecommunications sector, thereby effectively reducing disputes and delays.

3. Literature Review

This study focuses on the current 5G and upcoming 6G requirements outlined in widely cited research papers, where blockchain technologies are expected to play a crucial role in the future. Based on these requirements, interoperability is one of the most essential factors for connectivity in 6G 3D networks. These flexible 3D connections enable users to connect seamlessly with air, space, and ground service providers. Blockchain interoperability can effectively address multi-level interconnections. Previous research, including projects based on permissionless cryptocurrency solutions, has shown that such approaches are unsuitable for enterprise domains due to their high security and privacy expectations. The telecommunications domain is even more stringent due to the intense regulatory requirements it faces. Therefore, this study focuses solely on solutions suitable for telecoms within the permissioned blockchain space and explicitly does not aim to provide comprehensive coverage of the literature, as there is already extensive research on interoperability and its coverage.

This literature review primarily focuses on a thorough examination of significant research databases, including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect (Elsevier), Scopus, MDPI, IEEE Access, and arXiv, to identify relevant manuscripts related to blockchain interoperability and telecommunications in the context of blockchain technology. The citations for these selected works have been meticulously verified through Google Scholar. Instead of attempting a comprehensive overview of all relevant studies, this review strategically selects the most widely recognized and highly cited research addressing these solutions, along with projects developed by established organizations, such as International Business Machines (IBM). This review covers the most critical design categories and architectural approaches from these projects, research papers, and the literature in general. The scope of this study enables us to demonstrate research directions for the most substantial and suitable solutions using these categories and

architectural approaches, including research directions relating to the development of an ideal architecture, particularly for 6G communications. Finally, the carefully selected and reviewed solutions are compared to finalize the review and provide research directions through the selection of the most suitable solution. Finally, the chosen solution is verified to determine whether it meets the expected criteria for an ideal permissioned interoperability solution. Researchers reading this paper should consider this selection as a guide for future research on blockchain interoperability in the telecommunications field.

Zakari et al. [43] published a literature review focused on the opportunities and challenges of applying blockchain technologies in the pharmaceutical industry. In particular, they discussed five opportunities that blockchain technologies offer: transparency, reduced transaction time, security, cost efficiency, and irreversible transactions. Furthermore, five primary challenges associated with current blockchain technologies were identified, emphasizing the need for solutions to unlock these opportunities: interoperability, scalability, storage, social acceptance, and standardization requirements. These challenges intersect with known issues in the telecom industry, where interoperability is a critical hurdle. Zakari et al. highlighted the importance of security, which is essential for telecom cross-chain communication. Achieving technical interoperability does not eliminate security concerns, which remains a complex issue requiring targeted solutions. In cross-chain communication (CCC), each blockchain enforces separate access policies to maintain privacy and security across different telecom network operators. This introduces complex security requirements, posing a critical challenge to interoperability. Zamyatin et al. [44] have suggested that true interoperability requires a trusted third party, whereas many solutions aim to eliminate the need for intermediaries, aligning with the principle of decentralization.

Blockchain interoperability is a popular topic in supply chain tracking. A pivotal survey by Bellavista et al. [45] examined interoperable blockchains in collaborative manufacturing, which are integral to supply chains. They acknowledged that interoperability remains challenging despite numerous solutions, primarily due to ongoing security concerns. The authors concluded that the design of an interoperability solution must meet high-security expectations. In their survey, Bellavista et al. discussed a security solution called a trusted execution environment (TEE). Using this solution, Bellavista et al. proposed a relay scheme architecture in which information is passed through a relay layer, as shown in Figure 6. The TEE solution is designed to eliminate intermediary architectures and modules, facilitating interoperability between two distinct blockchains through a direct connection established by the relay component of each blockchain network. Such a TEE deployment with a relay architecture is feasible for two interoperated blockchains; however, the implementation can become complex if the end-to-end enterprise flow involves more than two blockchains. This could offer security and access rights to all interoperated blockchains but may quickly introduce access policy concerns and conflicts with independently operating blockchains. This is an evident concern regarding tamper-proof security and the strict access policy requirements in the telecom domain.

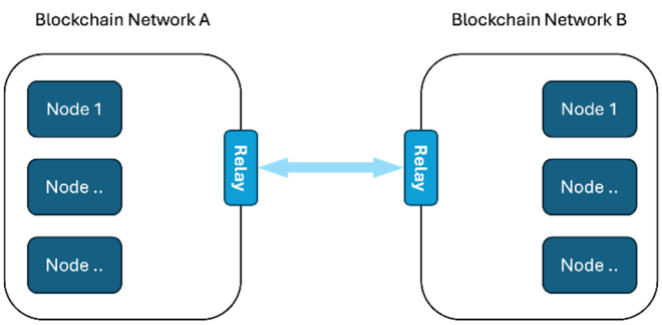

Figure 6. Relay scheme from Bellavista et al.

3.1. Interoperability Techniques

Exploring various popular descriptions, use cases, and interoperability techniques in this research field is essential for a better understanding of interoperability. According to Belchior et al. [46], interoperability is "the ability of a source blockchain to change the state of a target blockchain, enabled by cross-chain or cross-blockchain transactions, spanning across a composition of homogeneous or heterogeneous blockchain systems". This definition highlights that even though telecom operators tend to opt for standard and widely used private blockchain solutions, achieving homogeneity in blockchain networks remains challenging. Therefore, an ultimate solution should support diverse types of private blockchain networks.

Ren et al. [6] defined blockchain interoperability as "the ability to flexibly transfer assets, share data, and invoke smart contracts across a mix of public, private, and consortium blockchains without any changes to underlying blockchain systems". They further expanded the definition of the "ability to change the state of target blockchain" from Belchior et al. according to three essential features:

- Asset transfer;
- Smart contract execution;
- Data sharing.

These are the most common types of interoperability. Asset transfers and swapping are the most popular public cryptocurrency-based interoperability solutions. These use cases can also be applied to private blockchains, providing more flexibility in the telecom domain.

Vitalik Buterin, the founder of Ethereum [47], described the general concept of interoperability techniques in 2016. Buterin discussed three scenarios:

- Centralized or multi-signature notary schemes; • Sidechains/relays;
- Hash-locking.

These initial definitions are necessary to explore and expand our understanding of future interoperability schemas. According to Ren et al. [6], in 2023, interoperability techniques were categorized into five schemes:

- Notary Schemes: Consider trusted parties for centralized and decentralized exchanges. Notary schemes are trendy in cryptocurrency, as there is a high demand for the exchange of cryptocurrency coins worldwide. A famous example is the largest cryptocurrency exchange platform, Binance [48]. Binance is considered a trusted party when users require currency exchange. Notary schemes, such as trusted intermediaries, are considered centralized solutions and are not candidates for a final solution.
- Hashed Time Lock Contracts (HTLCs): Automatically swap currencies for permissionless networks. Unlike notary schemes, which require a trusted third party, the HTLC protocol automatically swaps currencies between cryptocurrency-based blockchains. There is no intermediary for an HTLC-based solution, which makes an HTLC a decentralized solution.
- Relay Schemes: Carry out transactions by transmitting them across blockchains, which can be either trusted or trustless. Trustless schemes agree on the communication protocols between the two blockchains and do not require trustees. If a relay scheme is built as a trusted relay, the solution is an intermediary-based centralized solution. The trustless relay approach is an ideal candidate if intermediaries need to be prevented.
- Blockchain-Agnostic Protocols: Build an abstraction layer to communicate and interoperate among diverse blockchains. Building an abstraction layer is essential for defining the solution as centralized or decentralized. In 2019, Abebe et al. [49] proposed a relay component that abstracts the existing blockchain modules from the interoperability layer. The architecture presented in [49] is constructed as a decentralized solution without a trustee and is one of the earliest solutions for permissioned blockchain networks built with Hyperledger [50,51].

The design offered by Abebe et al. also provides a side-by-side architecture and serves as a foundation for future projects using Hyperledger.

- Sidechains: Transfer assets from the mainchain to the sidechain. Consequently, the sidechain transfers the assets to the final chain. If this intermediary chain is responsible for the interoperability of two independent blockchains, the challenge lies in who owns the sidechain and how it is managed. Therefore, sidechains cannot act as a decentralized interoperability approach when deployed as intermediaries between interoperating blockchains. In the final solution, a sidechain can still be used if it is deployed as a side component of that chain and does not act as an intermediary.

As previously discussed, the five schemes presented primarily concentrate on the interoperability between two blockchain networks. However, when the interoperability extends to encompass three or more blockchain systems, the complexity of the process significantly increases. This complexity is particularly relevant in scenarios such as endto-end enterprise processes that traverse multiple blockchain environments. Belchior et al. called this scenario type a blockchain of blockchains (BoB) [46]. Figure 7 illustrates an international roaming use case where a decentralized, intermediary-free process is developed with blockchain interoperability. As seen on the left side of Figure 7, most telecom operators use clearing houses to facilitate the reconciliation of roaming records for their users in the visiting country. By implementing a direct interoperability architecture, clearing houses can be eliminated, as illustrated on the right side of Figure 7. This scenario can be viewed as a unified process between two or more operators. Once the blockchain networks are established for direct interoperation among multiple telecom operators, the process can be described as a BoB.

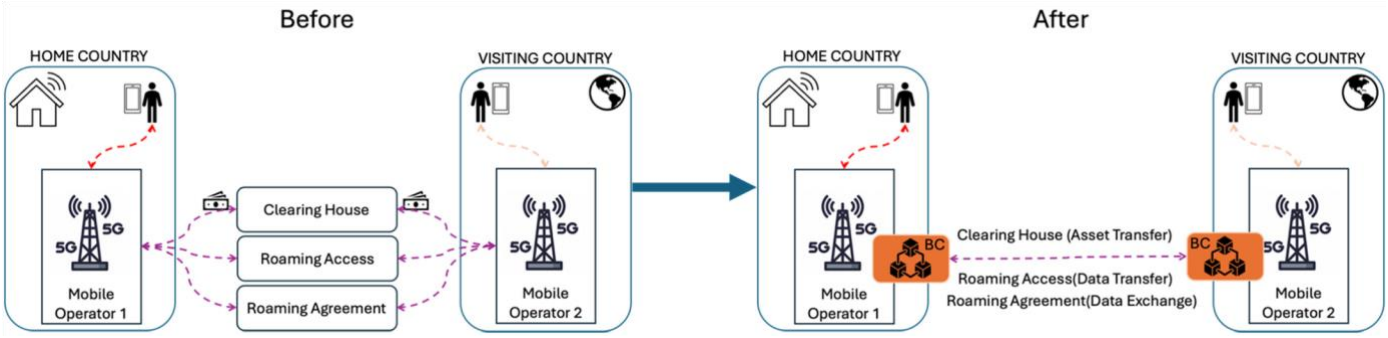

Figure 7. New international roaming process with blockchain interoperability. BC, blockchain.

Figure 7 illustrates how future telecoms can form interoperable processes with other telecom operators. Blockchain interoperability cases are also applicable to 6G. Figure 8 illustrates how different parties are integrated with the 3D 6G space for future telecom interoperation landscapes.

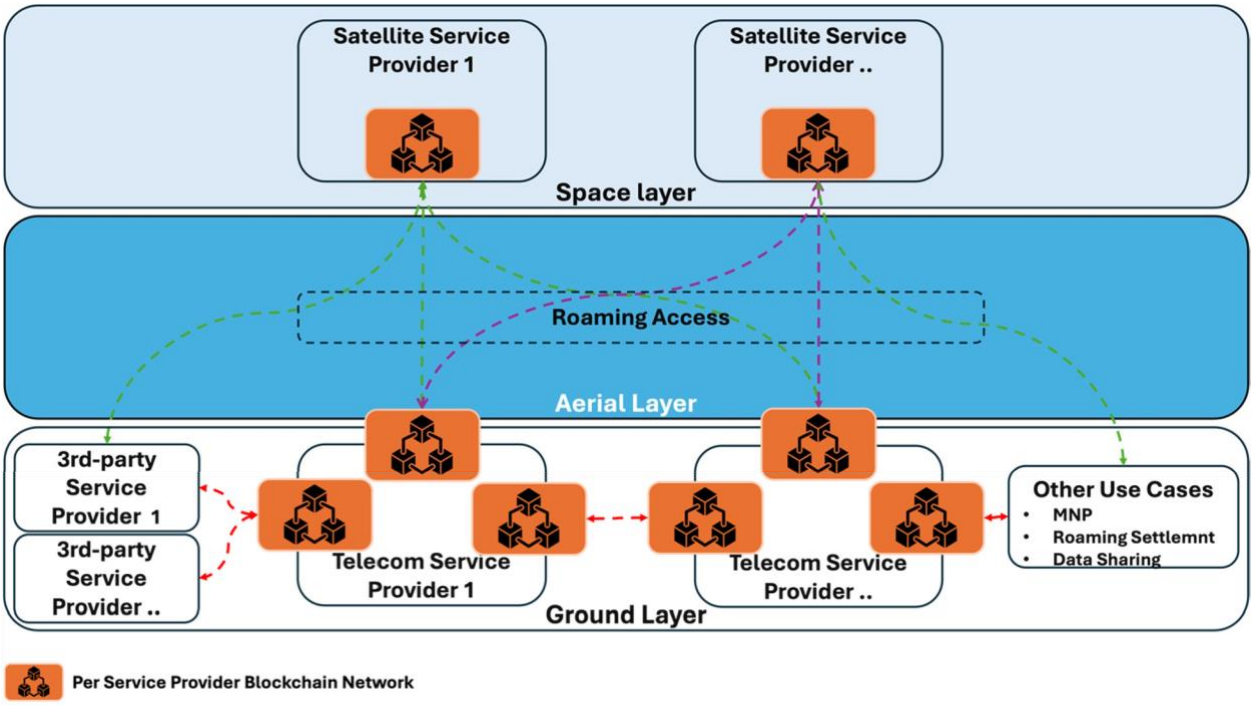

Figure 8. Blockchain interoperability service architecture for 6G Mobile Networks.

Figure 9 illustrates the mobile number portability process for three mobile operators. Once the MNP provider shown in Figure 5 is eliminated and handled as a decentralized process with no intermediary, the blockchain layers form a process between the three mobile operators.

For 5G and 6G networks, a decentralized MNP process can be established using interoperability layers. When a user triggers a porting request, the blockchain network sends the request to the target mobile subscriber. Suppose a user ports from Network 1 to Network 2, and their unique subscriber code (USC) is transferred to the target operator. Once the porting request is completed, asset exchange occurs from the blockchain perspective. Consequently, the new owner of the USC is shared with Operator 3. This is considered data sharing from a blockchain perspective, and an update is reflected in the blockchain's distributed ledger.

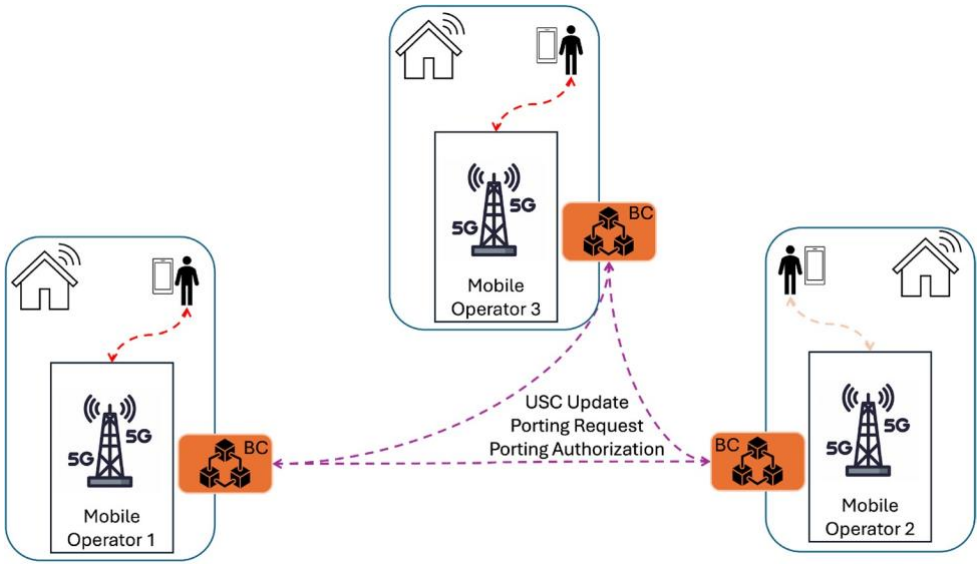

Figure 9. Mobile number portability porting process with blockchain interoperability.

In 6G, more enhanced interoperability use cases can be deployed for network slicing and spectrum sharing. Unlike the use cases presented in the literature that rely on a single blockchain

or intermediary, a blockchain interoperability architecture can be established for each service provider. This allows for much faster deployment of network slices by external service providers. As shown in Figure 10, the intermediary is replaced by standalone blockchains for each service provider, as illustrated in Figure 10.

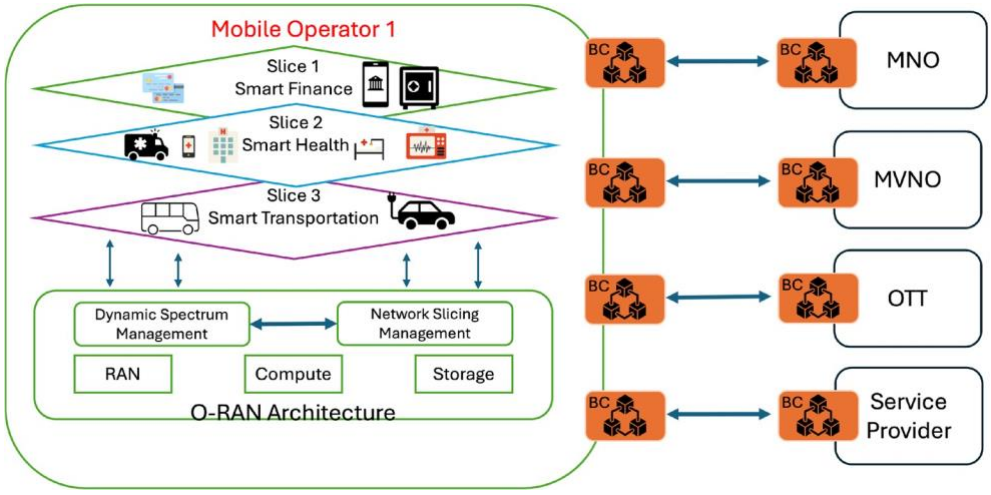

Figure 10. Blockchain interoperability architecture for network slicing and dynamic spectrum allocation (DSA).

The architecture in Figure 10 employs an O-RAN architecture, providing a flexible virtual resource allocation environment that permits the dynamic use of the available spectrum in constructing dedicated network slices requested by external parties. Utilizing smart contracts, spectrum allocation and network slicing rules can be established and executed in real time for each external service provider. This architecture delivers a privacypreserving interoperation environment and enhances performance by eliminating the intermediary broker for all external parties. With this approach, sensitive data remain on each side of the blockchain.

3.2. Future Telecom Use Cases with Blockchain Interoperability

Figure 8 shows two telecom providers with three blockchains deployed in their infrastructure. Each of these blockchains interacts with third parties or service providers. Based on the initial use cases, blockchain interoperability can be achieved in the following scenarios:

- National/International Roaming Access: Under agreed-upon conditions, a telecom service provider can connect to another national/international service/telecom provider, allowing users to use their roaming services seamlessly. Figure 6 illustrates this process.
- Roaming Payment Settlement: Without a roaming clearing house (intermediary), telecom providers can agree to pay to access customers' roaming records and settle payments directly with interoperating telecom providers.
- Mobile Number Portability: This can be introduced in countries where multiple telecom providers offer services, and users can port their services to other operators. However, this process currently utilizes trusted intermediaries managing the endto-end process, causing delays. Sixth-generation technology is expected to offer broader services than 4G and 5G. Using 6G with new-generation mobile phones possessing eSIM (embedded SIM) capabilities can boost the need for MNP with almost no operational difficulties. Once blockchain interoperability is deployed in each operator, the layer can eliminate the intermediaries and accelerate the end-to-end process. The new process is illustrated in Figure 7.
- Third-Party Service Access and Payment Settlement: Telecom operators use third-party content services. Blockchain interoperability can automate access to such services and content. In this line, third parties can access operators' blockchain layers to offer services, and payments can be made automatically using blockchain interoperability layers.

- Integrated Sensing and Communications for Roaming Devices: A standalone blockchain integrated with decentralized DID components and ZKP ensures a privacypreserving and tamper-proof architecture that eliminates eavesdroppers and external attacks. For national and international roaming, a side-by-side interoperability architecture enables seamless and secure connections of visitor devices by preventing unauthorized access and man-in-the-middle attacks.
- Networking Slicing and Dynamic Resource Sharing for External Operators: As shown in Figure 10, external parties and interconnected partners can access a mobile operator's resources using an interoperability architecture. A mobile operator may also conduct an auction if multiple parties are interested in utilizing network resources and spectra. Furthermore, a mobile operator can implement a blockchain-based crowd spectrum sensing model [52] in order to identify unused parts of the spectrum and offer the available spectrum to external parties through interoperability.

3.3. Essential Design Requirements

The blockchain-based interoperability architecture can cover the aforementioned use cases for future 6G telecoms. The essential design requirements for future telecoms are outlined below, based on the existing literature and popular projects:

- Single Point of Failure: According to Bhatia et al. [53], intermediaries such as notary schemes and sidechains introduce an SPoF into interoperability designs. In such a design approach, the interoperability of cross-chain communication is not operational when the intermediary goes down. This is a typical SPoF issue, as no alternative or backup solution exists.
- Decentralization: This is the most famous feature of blockchain technology. Blockchain promises to eliminate any authority or intermediary that may rule the interoperability process and make decisions when executing transactions. Atzori [54] discussed government-owned processes and asked whether such intermediary processes are necessary. As asserted in [54], blockchain technologies represent a disintermediation process that can remove intermediary state-owned processes. Atzori even mentioned the state or governing authorities causing an SPoF, as they cannot necessarily respond to the rapidly changing needs of society and face scalability challenges in delivering services. According to Chen et al. [55], decentralized enterprise models create new opportunities in this context. Similarly, Zheng et al. [56] indicated that blockchain cryptocurrencies' existing mining consensus models allow large miners to dominate the mining process. The authors of both studies suggested that centralized architectures or models be eliminated. As discussed in [38], the third frequency range of sixth-generation (6G) communication systems (FR3) requires meticulous channel modeling. This requirement is critical for the effective use and sharing of the available spectrum. If dynamic and agile spectrum sharing is centralized through a process established by a regulatory authority and facilitated by an intermediary application, this centralization may hinder the efficiency of spectrum sharing and introduce delays into the process.
- Safety of Access Management and Policies: Every independent permissioned blockchain has specific user-access management rules and policies. The rules, such as consensus mechanisms and executed smart contract-based rules, are particular to the blockchain network. When an interoperability architecture is offered, there is an expected tendency to synchronize and unify both blockchains' access management functions. Dagher et al. [57] have proposed a framework that preserves patient privacy in a national health system while using blockchain to manage private patient data access. According to Dagher et al., patient data confidentiality must be maintained if external access is required. Ren et al. [6] stated that the user identity and data can be detected when a trusted intermediary is used for interoperability. Therefore, intermediaries should be avoided, and trustless decentralized approaches should be chosen to keep users' identities safe.

- Sovereignty: In contrast to these approaches, Ghosh et al. [58] provided a crossnetwork identity framework that manages DIDs in an interoperability network. The self-sovereignty of blockchains in a multi-network architecture allows each blockchain to choose which users and groups can access their distributed ledger technologies (DLTs) from other chains and which functions they can execute through cross-chain communications. DIDs provide users with self-sovereignty. However, unified access management and policies for interoperated blockchains can introduce unexpected access leakages. These statements indicate that the user rights, access policies, and data are preserved and isolated. In such cases, DIDs with minimum user extensions can have much more secure and minimally affected interoperability.
- Core Blockchain Process Isolation: Pongnumkul et al. [59] have evaluated the performance of private blockchains under various workloads. Their results showed that the performance of blockchain frameworks still needs to be improved when high workloads are involved, and their processing speeds are not competitive with existing database systems. In their project, Ethereum and Hyperledger Fabric were compared, with Hyperledger Fabric emerging as the clear winner, exhibiting as much as ten times lower latency and much higher processing capacities. These results indicate that performance issues are critical in core blockchain transaction processing. Further performance drawbacks may be introduced when an additional interoperability solution is deployed in the core blockchain network. dApps and smart contracts are designed to facilitate interactions between blockchains and external applications. Vacca et al. [60] stated that the performance of a dApp is vital for assessing blockchain efficiency. Therefore, Vacca et al. evaluated the performance of both dApps and smart contracts

  in their research paper. They demonstrated the test results for dApp and smart contract deployments, which revealed varying performance outcomes. Any additional design frameworks for the core blockchain must be carefully conceived and tested before its implementation for interoperability. Such a design must also be isolated from the core blockchain process in order to minimize its impact on performance.
- Storage of Traceability Transactions: Once the number of transactions in a traditional blockchain increases, the long-term storage of transactions for traceability needs might become a real challenge. Musungate et al. [61] discussed using DLTs as a storage mechanism in main blockchain networks, which can quickly become large due to crowded blockchain domains. Traditionally, the mainchain has been expected to provide decentralization, intercommunication, security, and privacy. Therefore, features such as extensive storage management are focal points in most projects. According to Yadav et al. [62], these significant storage needs can quickly become a performance issue when the mainchain is queried frequently for transaction history. Therefore, they offered a storage land registry for national health systems, which are frequently queried by doctors and health system users. Musungate and Yadav suggested using a sidechain approach to achieve better performance and storage management, considering the mainchain only for the standard blockchain features. In an ideal interoperability design, significant storage needs and high-performance expectations can be addressed using sidechains instead of main-chain-based approaches.

After conducting a comprehensive literature review, the identified essential design requirements were crucial for developing an optimal blockchain interoperability project for future telecom domains. To achieve this, well-established projects from the existing literature and the research landscape were selected and reviewed in order to determine whether they could meet these essential design requirements. This approach enables telecom operators to accelerate the design and implementation of blockchain interoperability in current and future processes.

3.4. Review of Well-Known Permissioned Blockchain Interoperability Projects

Hyperledger Cactus [63] is the Hyperledger Foundation's initial interoperability project. It has been built as a bridge between Hyperledger Fabric and Ethereum for cross-chain communications. This was one of the earliest solutions implemented as a proofof-concept for permissioned blockchains, in which the cactus is considered a notary. Similar notary solutions built between two interoperated blockchains are classified as centralized solutions and intermediaries. This type of intermediary architecture also introduces an SPoF when an intermediary is unavailable.

Bradach et al. [64] proposed a gateway-based solution in 2022, where a gateway was used for cross-chain communications between two interoperating blockchains, namely Hyperledger and Corda. In the gateway part, a router is responsible for routing the message from the Hyperledger connector to the Corda connector. This approach can be considered a cross-chain trusted relay. Bradach et al. called this solution middleware, which is, by other means, an intermediary. As such, an intermediary solution is deployed outside both chains, it raises the question of who maintains this component. This approach generally breaks down decentralized architectures and presents a centralized solution.

Weaver [65] is a new approach that provides substantial enhancements compared with existing solutions. Weaver is based on Hyperledger Fabric and aims to provide interoperability between DLTs of the same or different types in permissioned blockchains. Weaver offers three use cases: data sharing, asset transfer, and asset exchange. Weaver introduces two modules as side components for each interoperating blockchain. The Intop is responsible for communicating with the core blockchain, while a relay is used to communicate with the interoperated blockchain and includes a communication module for each blockchain type (e.g., Corda, Hyperledger, and Ethereum). Weaver access management modules are currently being developed. Weaver's design is a side-by-side architecture and isolated framework, making it a decentralized solution. Weaver also uses asynchronous and message-based communication between interoperation modules. These two features make Weaver a more flexible and fault-tolerant solution. The fast and efficient communication protocol gRPC is also used to communicate between interoperated blockchains. The design choices were carefully made and could be essential for the provision of an ideal solution. It also aims to deploy a DID architecture for identity management. This design allows more than two blockchains to interoperate, making Weaver a strong candidate for the final permissioned blockchain interoperability solution.

Hyperledger Firefly [66] is a promising solution for the future of blockchain technologies. This project aims to accelerate the development time for future blockchains, notably through the introduction of a Web3 development framework. Firefly offers a solution based on the permissioned data passing through multiple chains. According to Kang et al. [67], Firefly requires a third party to validate the nodes, which introduces a security bottleneck. This type of third-party validation also introduces an intermediary that can be considered to be centralized. Although Firefly provides flexible development environment tools for dApps in Web3, its design architecture is very complex for simple data and asset transfers.

Hermes [68] was proposed by Belchior et al. and concentrates on fault tolerance for possible crash cases of middleware-based interoperability. Hermes aims to achieve an SPoF-free architecture to eliminate crash-causing cases and provide crash resilience. Hermes presented a valid use case that can be integrated into future designs. However, using a gateway-based intermediary architecture, particularly for a fault-tolerant design, renders Hermes a centralized architecture.

Yui [69] is an incubation project at Hyperledger Labs that enables cross-chain communication in heterogeneous blockchains. It uses the IBC protocol for cross-chain communications and offers a relayer service as middleware. Although the relayer cannot be considered a genuinely trusted party, it still acts as an intermediary module between the two interoperable blockchains. The initial

design does not support side-by-side deployment of the modules. However, if a side-by-side approach is applied in the later stages, it can also be considered a strong candidate.

Dinh et al. [70] proposed a design blueprint for interoperable blockchains, focusing on access control, cross-chain transactions, and communication as challenges that are valid for cross-chain telecommunications. Dinh et al. added access control, transactions, and communication to each side of interoperated blockchains, such as Weaver. These three components isolate the core blockchain from the interoperability modules. An interoperability architecture was proposed by Dinh et al. using side-by-side components. Their design renders the final architecture a decentralized solution through the use of a side-by-side approach and isolated modules. Bellavista et al. [45] proposed a similar side-by-side approach. The authors called their design a relay scheme, as it is responsible for transmitting transactions from one blockchain to another.

3.5. Comparison of Interoperability Projects

This review covered design issues and an analysis of well-known projects. As a result, an ideal telecom solution was proposed. This solution and approach can serve as a foundation for ensuring the interoperability of existing and future telecom projects. As previously demonstrated, permissioned blockchains include must-have features for future telecom and 6G deployment. Therefore, permissioned blockchains are a critical requirement in telecom use cases. In [70], the proposed initial design principles were presented with respect to supporting research papers, and details of these projects and design approaches were provided. If required, the readers of this paper can view the design architectures in [71], in which six design principles were covered: decentralization, homogeneous networks, smart contracts, wide range of contributors, SPoF, and side-by-side architecture. Here, these design principles are combined with the additional topics covered in this review, such as access management, the safety of policies, and core blockchain process isolation. Furthermore, in [71], heterogeneous blockchain interoperation was considered crucial for the enterprise processes formed by blockchain frameworks and telecom use cases.

This review's results established eight distinct comparison categories for an ideal blockchain interoperability solution, offering a forward-looking approach to the final design. These categories are defined as follows:

- Interoperability Type: Interoperability in blockchain solutions is characterized by various types, such as permissioned, permissionless, or hybrid. Each type of blockchain has its own feature set, which affects its security and access policies. These features are crucial in ensuring private blockchain interoperability in the telecom industry. The hybrid type involves a combination of private and public blockchains within a solution.
- Decentralization: Existing project designs include intermediaries, notaries, and sideby-side modules that facilitate interactions between blockchains. However, these patterns are sometimes decentralized. In the telecommunications field, intermediaries causing centralization should be removed.
- Multi-Network Support: Many solutions are tailored to connect two blockchains. Assessing whether a solution is scalable for future research and enterprise endeavors— such as multi-network integration—is crucial. These solutions should facilitate the interoperability of three or more blockchains, which is becoming increasingly important with the emergence of 6G technology.
- Chain Isolation: Isolating the mainchain provides a strict access policy and minimizes external effects. Most solutions offer direct access to a relay or an intermediary. This approach renders the solution vulnerable to attackers, particularly in the case of regulated telecoms. A thin interoperability layer with multiple layers has a minimal effect on the operation of the mainchain. It protects the solution from external attacks and provides high-level user privacy and security.

- Sovereignty: Numerous solutions presented in existing research projects provide integrated and synchronized access policies between interconnected blockchains. However, synchronizing access policies can create security vulnerabilities that hackers can exploit. The mainchain is expected to remain autonomous and enforce restricted minimal access policies for interoperability solutions without unifying the access policy with interconnected chains. This approach aims to safeguard user privacy and establish self-sovereignty in telecommunication.
- Storage Efficiency: In a traditional independent blockchain network, the mainchain is expected to be the central storage for status updates in DLTs. For interoperated blockchains, especially in the 6G context, the storage requirement is expected to be much higher due to the additional storage required for interoperability. According to Yang et al. [72], sidechains extend the storage capability of blockchains and are a better choice in certain blockchain use cases. Therefore, managing storage in an interoperational architecture is critical. According to the test results in [35], a sidechainbased solution eliminates the storage problem of the mainchain. The results from [72] also indicated that when data are separated into main- and sidechains, access to a separated sidechain is much more secure and has less of a performance effect on the mainchain.
- Support Community: If a larger developer community supports the selected solution and its components, it will have a much longer life cycle and can be considered a strong candidate for future projects. The strength of the support community for integrated blockchain networks and supported solutions is critical.
- Side-by-Side Design: Unlike the intermediaries proposed in most solutions, the sideby-side approach in each blockchain network strengthens sovereignty and chain isolation in interoperability.

4. Results and Discussion

The comparison and evaluation results of interoperability solutions are displayed in Table 1. Based on the requirements and essential feature set, researchers can identify the most suitable solution for a specific telecom use case.

Table 1. Comparison of interoperability solutions.

	Interoperability	Decentralized	Multi-Network	Mainchain Isolation	Sovereignty	Storage Efficiency	Side-by-Side Design	Support Community
Cactus [63]	Hybrid	×	×	×	×	×	×	Medium
Bradach et al. [64]	Permissioned	√×	√×	×	√×	×	×	Weak
Firefly [66]	Hybrid			×		×	×	Strong
Hermes [68]	Permissioned	×	√×	×	√×	× ×	×	Weak
Yui [69]	Hybrid	√×		×		×	√×	Medium
Dinh et al. [70]	Permissioned	√	×	√×	√×	×	√	Weak
Bellavista et al. [45]	Permissioned	√	×	√ √	√ √	×	√	Weak
Abebe et al. [49]	Permissioned	√	√×			×	√	Weak
Weaver [65]	Hybrid							Medium

To align with the strict privacy and security requirements of telecoms, only permissioned blockchain solutions should be considered for the final interoperability deployment. Furthermore, decentralization is crucial as intermediaries cannot be part of the final design. Mainchain isolation is a vital aspect that should be considered for improved performance and security; however, this may not be a high priority for an initial MVP (minimum viable product). Sovereignty is also a key deciding factor for privacy expectations, and the architecture should not be extended to a hybrid access management model. Due to the limited literature, a performance design solution can be planned for future research projects, as much of the existing literature has failed to provide solid arguments in this regard. A solution supported by a strong community will encourage researchers to invest in a futureoriented design. A side-by-side design is the most critical feature for decentralization and privacy security. The results indicate that Weaver is a comprehensive solution

for deploying a private multi-domain blockchain architecture. Weaver is also designed as a side-by-side architecture that supports multiple deployments of interoperated chains. In addition, the results of this study suggest that storage efficiency features are a critical consideration with respect to performance.

The Weaver architecture, as illustrated in Figure 11, was designed for heterogeneous hybrid blockchain networks. Weaver has recently been merged into the Hyperledger Cacti [73] project, with the aim of providing more cross-chain communication options by unifying the existing Cactus project with Weaver. As of January 2025, Weaver's design offers interoperability relays for Hyperledger Fabric, Hyperledger Besu [74], and R3 Corda [75], including data transfer, asset transfer, and asset exchange features. Hyperledger Besu is an Ethereum client designed as an enterprise-friendly blockchain supporting permissionless and permissioned requirements. R3 Corda is designed for financial service interoperability, allowing value exchanges between R3 users. These three projects are popular in research and are supported by a significant community.

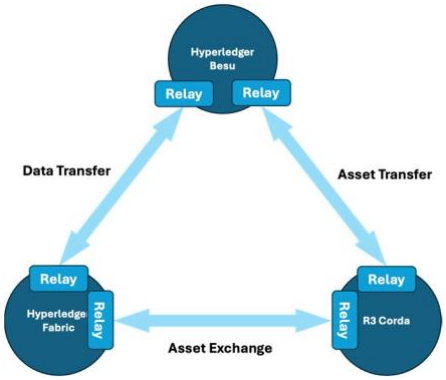

Figure 11. Weaver interoperability framework.

Due to its extensive feature coverage, Weaver's architecture is a strong candidate for future telecom use cases and 6G roaming/interconnection requirements. However, it should be noted that research alternatives have not yet addressed the requirements relating to performance and storage needs.

4.1. Challenges and Risks of Blockchain Interoperability in Telecommunications

Despite their potential benefits, there are challenges and risks associated with blockchain deployment in telecommunications. According to [24], six significant challenges can be introduced when using blockchain-based solutions: scalability; the measurement of decentrality; security and privacy; consensus algorithms; standards, policies, and legal issues; and interoperability in deploying blockchain in 6G networks. According to the existing literature, interoperability has the following sub-challenges:

- Technical complexity: Implementing interoperability across multiple blockchain networks requires technical expertise and compatibility among protocols, blockchain networks, and consensus algorithms.
- User Privacy and Separated Identity Management: When two or more standalone blockchain networks are expected to interoperate, the user identities in each network must remain private and undiscoverable by other interoperated networks.
- Scalability: Blockchain networks—particularly public permissionless blockchains such as Bitcoin and Ethereum—already face scalability issues with a high volume of transactions. Such problems can also occur in the telecommunications industry, even if permissioned blockchains are used. Implementing blockchain interoperability in the telecommunications industry is a complex task requiring technical expertise and compatibility between different protocols and consensus algorithms. It involves collaboration between multiple service

providers and establishing common standards to ensure seamless integration and communication; however, it can also introduce performance and scalability issues. Using homogeneous blockchains for interoperation has a high potential for addressing performance and fewer scalability issues.

- Regulatory challenges: Blockchain technology in the telecommunications sector may face regulatory hurdles, as it involves exchanging and storing sensitive customer data. Implementing blockchain interoperability in the telecommunications sector requires collaboration among service providers to establish common standards and overcome regulatory challenges surrounding data privacy and security. With the regulating body, telecom operators might need help with respect to planned blockchain-based interoperability deployment requirements. In addition to technical challenges, telecom sector leaders are expected to address regulatory issues by cooperating with regulatory bodies.

Considering these advantages and challenges, it is crucial to carefully choose the most suitable blockchain and interoperability solutions to effectively address a wide range of telecommunications scenarios, particularly in the context of 6G mobile networks. Given the telecom sector's emphasis on robust privacy and security measures, permissionless and public blockchain solutions do not apply to telecoms. Therefore, this study explores private and permissioned solutions and strategies. The following section presents a comprehensive review of the carefully selected literature in order to identify and evaluate ideal interoperability solutions and compare them to determine the most suitable candidates.

4.2. Contribution to the Research Space

The insights gained from this study and the proposed research directions contribute to the research domain in several ways. The importance of interoperability across multiple blockchain networks, especially in the context of 6G projects, is illustrated through a comparison of traditional interoperability between two distinct blockchain networks. The application of the Weaver framework indicates the feasibility of integrating two or more blockchain networks, serving any multi-network domain that requires a permissioned interoperability design. Furthermore, the critical role of security in telecommunications is highlighted by side-by-side architectures, which prevent unified access policies, thereby protecting the access layers of the mainchain. This is essential for ensuring telecom users' privacy and security while addressing some critical regulatory matters. This discussion also emphasizes the importance of primary chain isolation regarding blockchain interoperability. As the performance of a blockchain is vital, any extended design proposal must undergo thorough experimentation, mainly when introducing complex 6G interconnection features. Finally, this literature review demonstrates how blockchain's inherent characteristics, such as enhanced transparency and traceability, play a crucial role in interoperability and interconnection within telecommunications, particularly 6G 3D mobile networks. As the 6G 3D network incorporates complex access policies, robust access management tasks will be introduced exponentially. Implementing a unified access management strategy can reveal vulnerabilities to cyber-attacks. This study demonstrated that a side-by-side approach makes it feasible to avert cyber-attacks and establish a tamper-proof architectural framework, thereby ensuring end-to-end traceability and transparency.

5. Research Challenges and Future Directions

The findings from this study outline potential avenues for current and future applications of blockchain technology in the telecommunications sector. The analysis suggested the Weaver framework as a high-potential model for future research. Weaver's unique design, characterized by its isolated structure, is projected to significantly reduce the impacts on interoperability and traceability performance. To proceed with further research, it is essential to understand the key

challenges concerning interoperability that could impact the successful implementation of Weaver in the telecom research area.

5.1. Deployment Challenges

Weaver's high potential presents a couple of challenges. Weaver is currently under development, and some critical features still require discussion. Below are the essential topics that must be considered for its successful deployment:

- Scalability: The history of blockchain technology is full of discussions regarding performance issues. As discussed in the previous sections, new blockchain projects have been designed to address scalability and performance limitations, particularly in public blockchain research. Sixth-generation communications, characterized by seamless transitions between interoperated telecom operators, require strong performance due to ongoing transaction processing for service access and usage tracking. According to [76], 6G can meet the increasing demand for higher capabilities and broader-spectrum communications in mobile technologies. According to [77], once the Distributed Metaverse [77] is deployed in mobile networks, such an architecture will require higher scalability and capability. Gadekallu et al. [78] also discussed performance as a challenge for blockchain for edge-of-things (BEoT) devices deployed in 5G networks when the number of edge devices increases significantly. Therefore, they suggested moving algorithms to edge devices in order to minimize data travel between data centers and edge devices. Such a scenario will become more challenging when the data from these BEoT devices travel between 6G networks.
- Storage Management: When transaction processing per planned time frame increases significantly, storing these transactions on the mainchain can become a challenge and a performance bottleneck for end-to-end transaction storing and traceability. In [79], Xie et al. surveyed the scalability of blockchain systems. According to them, traditional blockchain nodes deployed for enterprise use require more storage space. If this issue is not addressed, transactional delays could occur as network density increases. In [80,81], off-site solutions were explored as alternatives to improve performance and make storage management more efficient. The authors of [80] also examined the TEE solution proposed by Bellavista et al. [45] as an off-chain option. This challenge necessitates further research into off-chain solutions, including sidechain options, as suggested in this study.
- Regulatory Hurdles: Despite various alternative interoperability solutions, regulatory bodies may permit information exchange between telecom providers only if an intermediary is utilized. In many countries, regulatory bodies designate official intermediaries for information exchange, as regulations and processes are necessary. Therefore, regulatory bodies must develop and implement new regulations for a direct peer-to-peer information exchange framework managed by a potential interoperability solution. Blockchain's immutability, encryption, and traceability features, combined with its tamper-proof architecture, can significantly help in persuading regulatory bodies to accept interoperability architectures. In this context, blockchain deployments are expected to provide regulatory bodies with access to trace spectrum allocation, mobile number porting, and roaming usage. Using well-established and widely supported interoperability solutions supported by the Linux Foundation (e.g., Hyperledger Fabric, Weaver) can streamline the certification processes initiated by regulatory bodies.

5.2. Future Directions

It is essential to focus on maintaining the three critical aspects of blockchain technology: decentralization, which spreads control across multiple locations or entities by eliminating intermediaries; immutability, which ensures that information cannot be altered once it is added;

and transparency, which allows for the clear visibility of transactions. This literature review highlights four additional areas for future research and discusses why they must be focused on:

- Side-by-Side Architectures: Weaver offers a side-by-side architecture that eliminates intermediaries. Future research on Weaver and similar approaches should be continued by observing future developments in such projects. Any new solutions are also expected to provide a side-by-side architecture to avoid intermediaries and centralized processes.
- High-Security-Oriented Approaches: Interoperability solutions risk introducing vulnerabilities when interoperated blockchain networks are combined with restricted access and privacy policies. Weaver's thin integration layer eliminates such merging

  and provides an ideal approach for enhancing security. However, any new design approach must be evaluated to determine whether it aligns with the strict access and privacy policy requirements.
- Isolated Mainchain: According to [72], there are three main advantages when a mainchain is isolated and integrated with a sidechain. A sidechain can be used to store and process use cases with high transaction requirements. It can also provide more space for transaction storage, faster data access, and higher security. The project for the sidechain proposed in [72] was conducted to safeguard customer privacy regarding medical information in the health sector. Such solutions, designed for customer privacy, can also be applicable to telecoms. A popular review paper [82] provided an in-depth analysis of existing sidechain solutions built for permissionless networks and examined four well-known sidechain solutions, including Loom [83]. This review paper can be considered for future sidechain research. According to [84], smart contracts are essential for deploying sidechain solutions. A smart contract-based layer must be developed and deployed if the mainchain requires integration with the sidechain. When further sidechain solutions, such as those in [80,85], are analyzed, it is evident that they use permissionless or cryptocurrency-based solutions, such as Ethereum, Ardor, or Loom, which lack strict privacy features. This indicates that more research is required on privately permissioned network-based sidechains. The solution in [86] integrates a sidechain with the mainchain using a cross-chain smart contract architecture. Interoperable networks must consider using smart contracts once the interoperability between heterogeneous networks is required. Considering the existing literature, this research area might bring potential advantages, especially for 5G and 6G interoperability, which requires higher performance and security.
- Scalable Architectures: As suggested in this study's final approach, scalable architectures such as sidechains require further analysis to determine their potential for contributing to the research space. In [81], the authors presented a sidechain approach for enhanced efficiency and faster transaction processing in 5G use cases. Most sidechain solutions aim to improve the mainchain's performance by minimizing interactions outside the blockchain network. This method is crucial for future research, and the selected requirements must retain the advantages of private blockchain architectures for an ideal interoperability framework.
- Performance: This literature review excludes the performance metrics of blockchain interoperability solutions. The proposed solution, Weaver, is constructed using components from Hyperledger Fabric 2.5. The two selected sources for performance appear to meet the general performance expectations for 6G. The test results presented in [87] indicated that Hyperledger Fabric with the Byzantine Fault-Tolerant consensus algorithm can achieve 3500 transactions per second (TPS) with a confirmation time of less than 1 s. The second source was announced by the Linux Foundation in February 2023. In [88], the performance results for version 2.5 ranged from 1500 to 2500 TPS across different setups. However, the Weaver components have not yet been tested with relay components in a side-by-side architecture.

Each telecom use case discussed in this paper requires end-to-end benchmarks in order to establish specific requirements for side-by-side architectures.

6. Conclusions

Blockchain interoperability remains a prominent area of future research, particularly in the telecommunications industry. It can significantly improve various aspects of the telecommunications infrastructure, especially in the context of the existing 5G and anticipated 6G 3D SAGIN designs. The integration of blockchain and its interoperability features is expected to provide significant benefits, including flexibility, immutability, security, efficiency, and transparency. Closely monitoring blockchain interoperability research is crucial to fully leverage these advantages. Numerous interoperability projects are currently available and continuously expanding, particularly in the field of permissioned blockchain research. However, the existing research is minimal and has not yet reached a universally applicable level. Based on the ongoing research and literature review findings, more time is required to develop a comprehensive solution for telecom interoperability. The emergence of the Metaverse is also a significant driver, as it aims to establish a fully decentralized architecture and will also be integrated into a telecom service provider network once deployed. The interoperability of permissioned and permissionless networks is essential for the Metaverse [68], making it a crucial focus of new and ongoing research projects. The results presented here allow for the identification of potential research directions for telecom interoperability and highlight essential design considerations as new research areas, allowing researchers to approach the topic from a broader perspective.

Author Contributions: Conceptualization, S.B.; methodology, S.B. and S.G.; software, S.B.; validation, S.B; formal analysis, S.B. and S.G.; investigation, S.B.; resources, S.B and S.G.; writing—original draft preparation, S.B.; writing—review and editing, S.B., S.G. and T.S.; visualization, S.B.; supervision, S.G. and T.S.; project administration, S.B. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Deng, Zhihong, Chunming Tang, Taotao Li, Parhat Abla, Qi Chen, Wei Liang, and Debiao He. "Enhancing Blockchain Cross Chain Interoperability: A Comprehensive Survey." arXiv preprint, May 2025. https://arxiv.org/abs/2505.04934.
2. Sarker, Md Forhad, and T. Hossain. "Blockchain-Enabled Cooperative Spectrum Sensing in 5G and B5G Wireless Networks." Journal of Network and Computer Applications 2024.
3. "Blockchain Interoperability in the Public Sector." SSRN, April 2025. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5242327.
4. Lee, Mingwu, and Sandra Richards. "Design and Implementation of a Decentralized Trustless Data Sharing System for Future 6G Telecom." Journal of Information Security and Applications 2025.
5. Wang, Jin, and Chong Liu. "A Trustless and Scalable Blockchain Interoperability Protocol for Multi-Chain Telecom Systems." ACM Proceedings on Blockchain Technology 2025. https://dl.acm.org/doi/10.1145/3696410.3714867.
6. "Bridging the Gap: Achieving Seamless Interoperability Between Ethereum-Based Blockchains." Software: Practice and Experience 2025. https://onlinelibrary.wiley.com/doi/10.1002/spe.70008.
7. Kaur, Manpreet, and David P. Taylor. "Blockchain Technology Meets 6G Wireless Networks: A Systematic Review." Journal of King Saud University - Computer and Information Sciences, 2024.
8. "A Survey on Advancements in Blockchain-Enabled Spectrum Access Mechanisms for 6G Networks." Nature Scientific Reports, December 2024. https://www.nature.com/articles/s41598-024-82126-y.
9. "Blockchain Interoperability: The Current State in 2025." CryptoEQ Report, March 2025. https://www.cryptoeq.io/articles/blockchain-interoperability-solutions.

10. "Blockchain Interoperability: Future of Seamless Integration." Rapid Innovation Blog, September 2024. https://www.rapidinnovation.io/post/blockchain-interoperability-bridging-diverse-networks-for-seamless-integration-2024.
11. Patrick, G. "Top 7 Blockchain in Telecom Companies Leading Interoperability and Innovation." Verified Market Research, July 2025.
12. "Blockchain Interoperability Market Report 2025 (Global Edition)." Cognitive Market Research, September 2024.
13. Chen, Z., Wu, Y., & Liu, Q. "Towards Blockchain Interoperability: A Comprehensive Survey on Cross-Chain Solutions." ScienceDirect, April 2025.
14. D. B. Mehta, “The Role of Artificial Intelligence in Optimizing Telecommunications Networks,” International Journal of Multidisciplinary Research and Publications, vol. 7, issue 11, pp. 65–71, 2025
15. Lee, M., Richards, S. "Design and Implementation of a Decentralized Trustless Data Sharing System for Future 6G Telecom." Journal of Information Security and Applications, 2025.
16. Wang, J., Liu, C. "A Trustless and Scalable Blockchain Interoperability Protocol for Multi-Chain Telecom Systems." ACM Proceedings on Blockchain Technology, April 2025.
17. Patrick, G. "Blockchain Interoperability for Future Telecoms." Sciety, March 2025.
18. "Blockchain In Telecom Market Share and Statistics - 2035." Fact.MR, December 2024.
19. "Web3 and Telecom: Blockchain's Role in the Industry." TeckNexus, May 2025.
20. "Blockchain Interoperability Market Size, Global Reports - 2034." Market Research Future, August 2025.
21. "Blockchain-based Inter-Operator Settlement System." ScienceDirect, 2025.
22. "Blockchain Interoperability for Enterprises in 2024." Rapid Innovation, September 2024.
23. "Blockchain Interoperability: The Current State in 2025." CryptoEQ Report, March 2025.
24. "Open RAN Enables Innovation and Flexibility for Future Networks." Comba USA, January 2022.
25. "OpenRAN - Telecom Infra Project." Telecom Infra Project, June 2025.
26. "Event Recap: Exploring Open RAN's Role in Transforming Telecom." Berkeley CLTC, April 2025.
27. "Open RAN: Five Defining Themes for 2021." GSMA Intelligence, February 2021.
28. Gkagkas, G., Vergados, D.J., Michalas, A., Dossis, M. "The Advantage of the 5G Network for Enhancing the Internet of Things and the Evolution of the 6G Network." Sensors, 2024.
29. Dangi, R., Choudhary, G., Dragoni, N., Lalwani, P., Khare, U., Kundu, S. "6G Mobile Networks: Key Technologies, Directions, and Advances." Telecom, 2023.
30. Kaur, M., Taylor, D.P. "Blockchain Technology Meets 6G Wireless Networks: A Systematic Review." J. King Saud Univ. - Comp. Inf. Sci., 2024.
31. Srivastava, A., Kumar, S., Rajput, N.S. "6G Wireless Communication Technology Operating at Frequencies Beyond 200 GHz: System-Level Modeling and Experimental Insights." African Journal of Biomedical Research, December 2024.
32. "A Survey on Advancements in Blockchain-Enabled Spectrum Access Mechanisms for 6G Networks." Nature Scientific Reports, December 2024.
33. Cosmos Protocol. "Cross-Chain Interoperability Report 2024." IBC, December 2023.
34. Sarker, M.F., Hossain, T. "Blockchain-Enabled Cooperative Spectrum Sensing in 5G and B5G Wireless Networks." Journal of Network and Computer Applications, 2024.
35. "Bridging the Gap: Achieving Seamless Interoperability Between Ethereum-Based Blockchains." Software: Practice and Experience, July 2025.
36. Ren, K., Ho, N.-M., Loghin, D., Nguyen, T.-T., Ooi, B.C., Ta, Q.-T., Zhu, F. "Interoperability in Blockchain: A Survey." IEEE Trans. Knowl. Data Eng., 2023.
37. Cuellar, D., Sallal, M., Williams, C. "BSM-6G: Blockchain-Based Dynamic Spectrum Management for 6G Networks: Addressing Interoperability and Scalability." IEEE Access, 2024.
38. Pasdar, A., Lee, Y.C., Dong, Z. "Connect API with Blockchain: A Survey on Blockchain Oracle Implementation." ACM Comput. Surv., 2023.

39. Ezzat, S.K., Saleh, Y.N.M., Abdel-Hamid, A.A. "Blockchain Oracles: State-of-the-Art and Research Directions." IEEE Access, 2022.
40. Raval, S. "Decentralized Applications: Harnessing Blockchain Technology." O'Reilly Media, 2016.
41. Park, A., Wilson, M., Robson, K., Demetis, D., Kietzmann, J. "Interoperability: Our Exciting and Terrifying Web3 Future." Business Horizons, 2023.
42. Qin, R., Ding, W., Li, J., Guan, S., Wang, G., Ren, Y., Qu, Z. "Web3-Based Decentralized Autonomous Organizations and Operations: Architectures, Models, and Mechanisms." IEEE Trans. Syst. Man Cybern. Syst., 2022.
43. Jabbar, S., Lloyd, H., Hammoudeh, M., Adebisi, B., Raza, U. "Blockchain-Enabled Supply Chain: Analysis, Challenges, and Future Directions." Multimed. Syst., 2020.
44. Kalla, A., De Alwis, C., Porambage, P., GVºr, G., Liyanage, M. "A Survey on the Use of Blockchain for Future 6G: Technical Aspects, Use Cases, Challenges and Research Directions." J. Ind. Inf. Integr., 2022.
45. Tataria, H., Shafi, M., Molisch, A.F., Dohler, M., SjVðland, H., Tufvesson, F. "6G Wireless Systems: Vision, Requirements, Challenges, Insights, and Opportunities." Proc. IEEE, 2021.
46. Chowdhury, M.Z., Shahjalal, M., Ahmed, S., Jang, Y.M. "6G Wireless Communication Systems: Applications, Requirements, Technologies, Challenges, and Research Directions." IEEE Open J. Commun. Soc., 2020.
47. Cui, H., Zhang, J., Geng, Y., Xiao, Z., Sun, T., Zhang, N., Liu, J., Wu, Q., Cao, X. "Space-Air-Ground Integrated Network (SAGIN) for 6G: Requirements, Architecture and Challenges." China Commun., 2022.
48. Ray, P.P. "A Review on 6G for Space-Air-Ground Integrated Network: Key Enablers, Open Challenges, and Future Direction." J. King Saud Univ.-Comput. Inf. Sci., 2022.
49. Chafii, M., Bariah, L., Muhaidat, S., Debbah, M. "Twelve Scientific Challenges for 6G: Rethinking the Foundations of Communications Theory." IEEE Commun. Surv. Tutor., 2023.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.